

Multi-Mode Heat Transfer

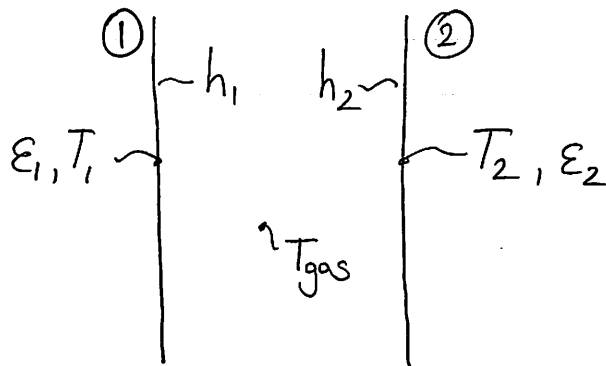
What if we have radiation & convection. How do we handle this?

Well, since:

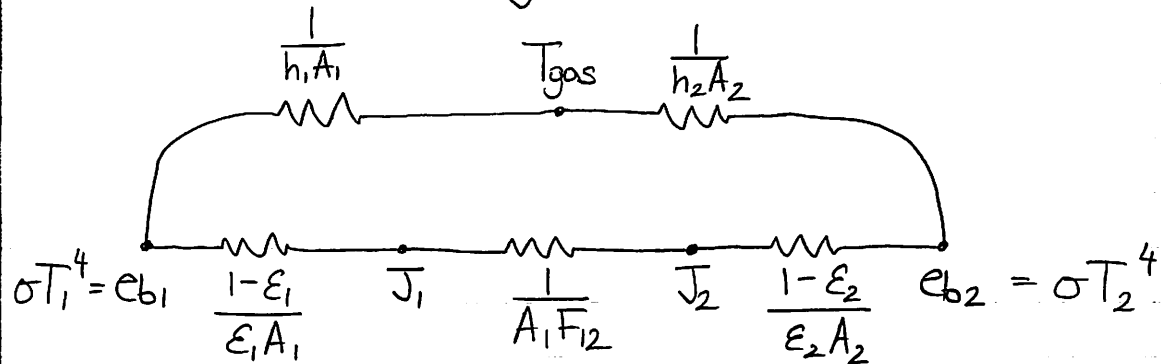
Radiation $\Rightarrow$ independent of medium
 Convection $\Rightarrow$ medium dependent

} Parallel paths.

For example:

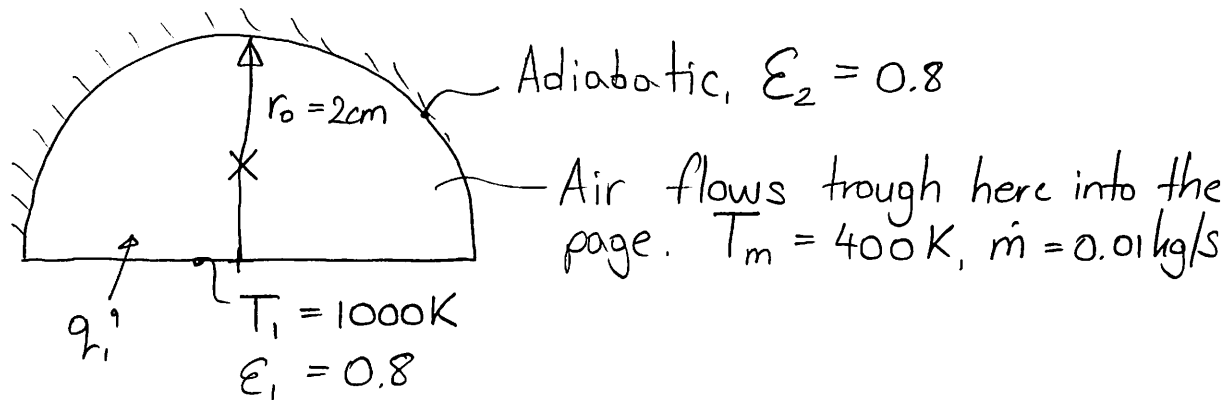


Our resistance diagram becomes:



We need to solve this complete resistance diagram and it usually involves iteration since T_{gas} is unknown.

Example | Consider an air heater : (semi-circular)



Find the temperature of the adiabatic surface, and the heat required per unit length (q'_1) to maintain $T_1 = 1000 \text{ K}$. Assuming fully developed internal flow:

Look up properties in Table A.4 or Incropera (Textbook)

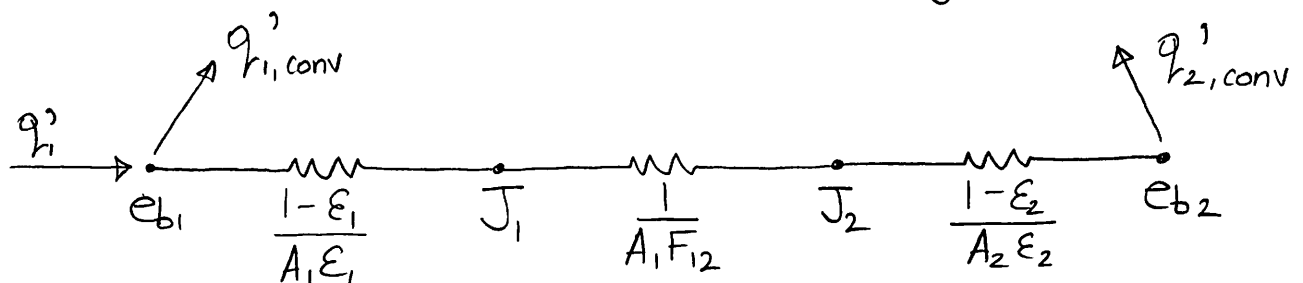
$$k_{\text{Air}} = 0.0338 \text{ W/m} \cdot \text{K}$$

$$\mu = 230 \times 10^{-7} \text{ kg/(s} \cdot \text{m)}$$

$$C_p = 1014 \text{ J/kg} \cdot \text{K}$$

$$Pr = 0.69$$

Let's draw our thermal resistance diagram:



Since surface 2 is insulated : $q'_{r2, \text{rad}} = q'_{r2, \text{conv}}$ ①

Net radiative transfer into 2. ⑦⑨

We can evaluate the radiative and convective balance on surface 2 as:

$$\frac{\sigma(T_1^4 - T_2^4)}{\frac{1-\epsilon_1}{A_1\epsilon_1} + \frac{1}{A_1F_{12}} + \frac{1-\epsilon_2}{A_2\epsilon_2}} = hA_2(T_2 - T_m) \quad \begin{array}{l} \downarrow \text{Air temperature} \\ T_m \end{array}$$

We know: $F_{12} = 1$ (doesn't see itself, $F_{11} = 0$)

$$A_1 = 2r_o \text{ (area per unit length)}$$

$$A_2 = \pi r_o \text{ (area per unit length)}$$

Now for the fluids problem: (convection)

$$Re_o = \frac{\rho u D_h}{\mu} = \frac{\dot{m} D_h}{A_c \mu} = \frac{\dot{m} D_h}{(\pi r_o^2 / 2) \mu}$$

$\hookrightarrow$ Cross sectional area

$$D_h = \frac{4A_c}{P} = \frac{2\pi r_o^2}{\pi(\pi+2)} = \frac{0.04\pi}{\pi+2} = 0.0244 \text{ m}$$

$$Re_o = \frac{(0.01 \text{ kg/s})(0.0244 \text{ m})}{(\pi/2)(0.02 \text{ m})^2 (230 \times 10^{-7} \text{ kg/m.s})} = 16900$$

So we have turbulent fully developed flow.

Flipping back to our correlation table for internal flow:
Table 8.4, pg 137 of notes.

We can use (8.60)^d $\Rightarrow$ Dittus-Boelter equation

$$Nu_0 = \frac{h D_h}{k_{air}} = 0.023 Re_D^{4/5} Pr^{0.4}$$

$$Nu_0 = 0.023 (16900)^{4/5} (0.69)^{0.4} = 47.8$$

$$h = \frac{k_{air} Nu_0}{D_h} = \frac{(0.0338 \text{ W/m}\cdot\text{K})(47.8)}{(0.0244 \text{ m})} = 66.2 \text{ W/m}^2\cdot\text{K}$$

Now we can work with our energy balance on surface 2.

$$\frac{\sigma(T_1^4 - T_2^4)}{\frac{1-\epsilon_1}{A_1\epsilon_1} + \frac{1}{A_1F_{12}} + \frac{1-\epsilon_2}{A_2\epsilon_2}} = h A_2 (T_2 - T_m) \Rightarrow \text{Divide through by } \begin{matrix} A_1 = 2r_o \\ A_2 = \pi r_o \end{matrix}$$

$$\frac{(5.67 \times 10^{-8})(1000^4 - T_2^4)}{\frac{1-0.8}{0.8} + 1 + \frac{1-0.8}{0.8} \left(\frac{2}{\pi}\right)} = (66.2) \left(\frac{\pi}{2}\right) (T_2 - 400)$$

Simplifying, we obtain:

$$(5.67 \times 10^{-8}) T_2^4 + 146.5 T_2 - 115313 = 0$$

Using iteration (guess T_2 & check solution), we obtain:

$$\boxed{T_2 = 696 \text{ K}}$$

Now we can solve for q_1' at the heated surface 1.

An energy balance reveals:

$$q_1' = q_{1,\text{rad}}' + q_{1,\text{conv}}' = q_{2,\text{conv}}' + q_{1,\text{conv}}'$$

All heat in (q_1') is taken away by convection. (181)

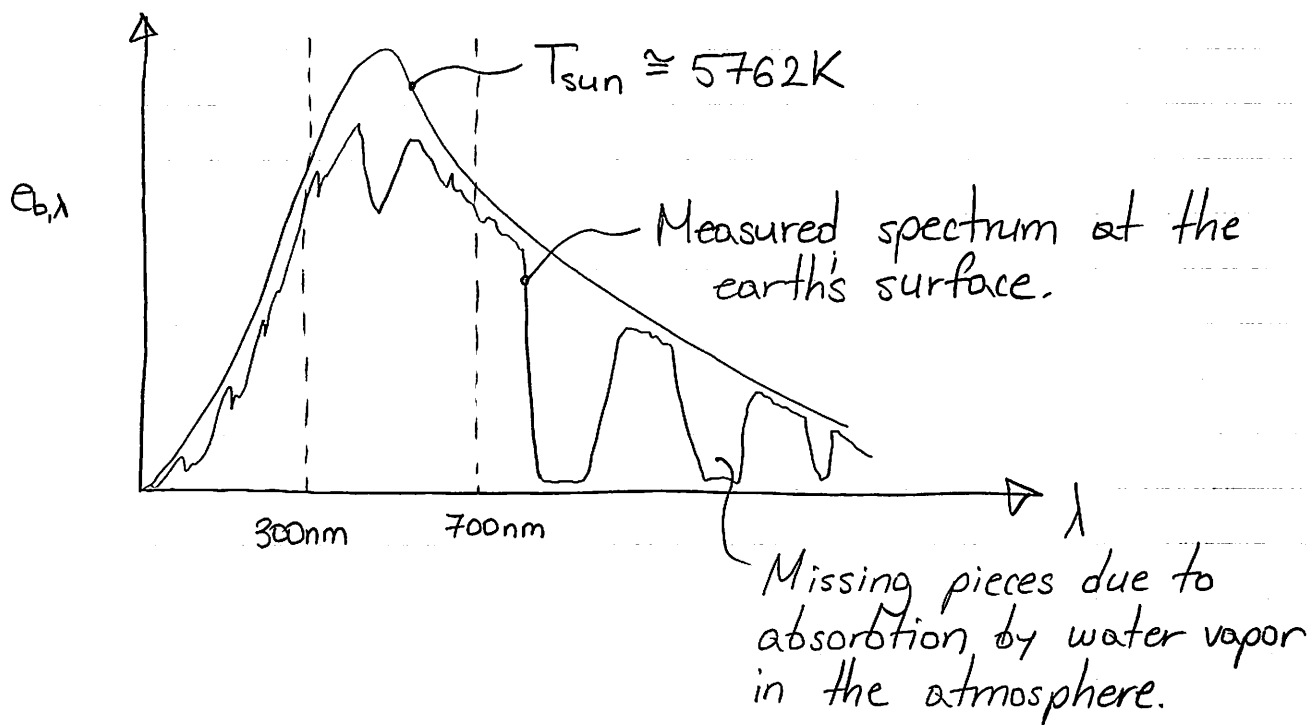
So on a per unit length basis:

$$q'_1 = h(\pi r_o)(T_2 - T_m) + h(2r_o)(T_1 - T_m)$$

$$= 66.2 \left[(\pi(0.02\text{m}))(696 - 400) + 2(0.02\text{m})(1000 - 400) \right]$$

$$q'_1 = 2820 \text{ W/m}$$

Solar Radiation



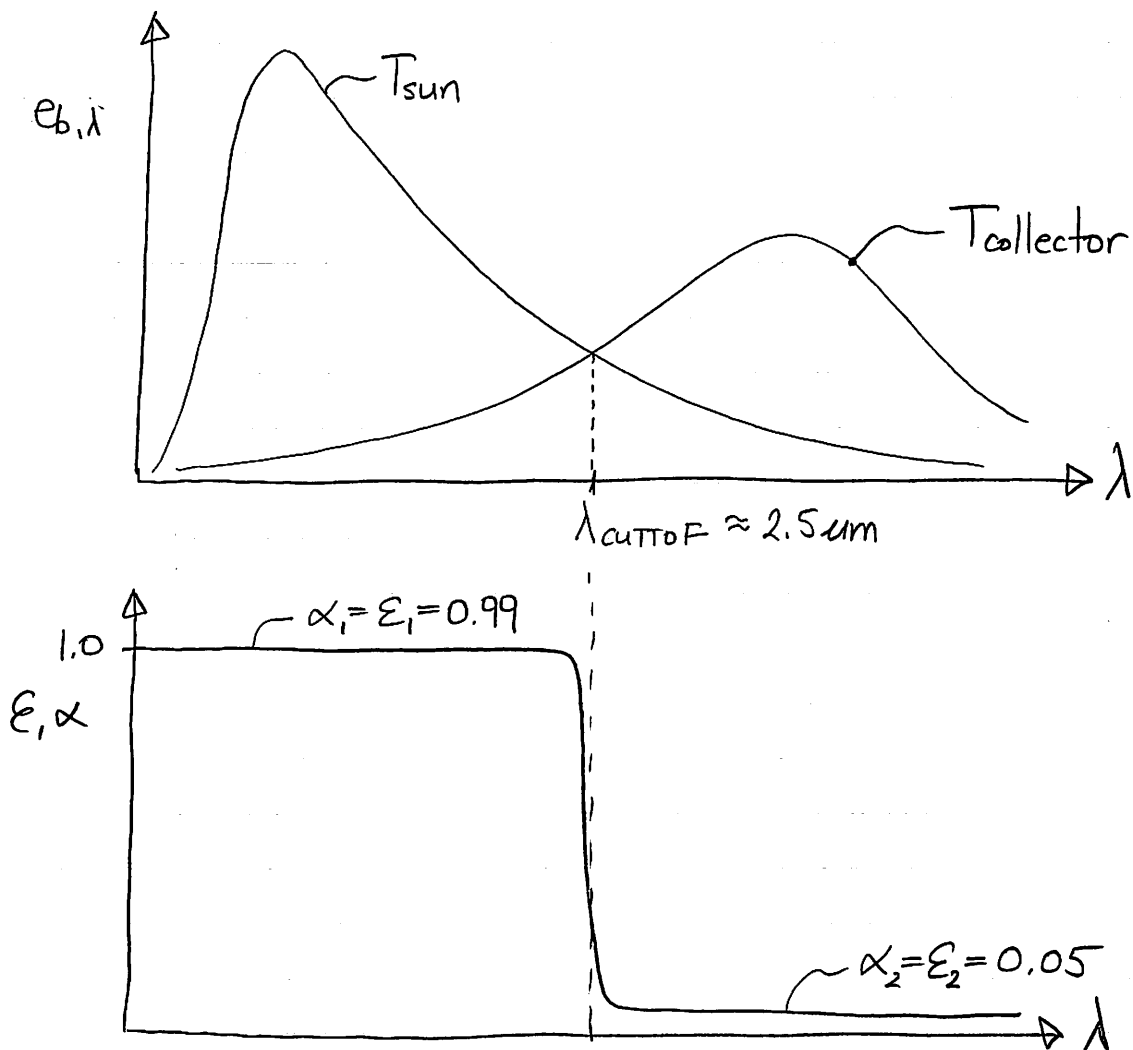
$$q''_{\text{sun}} = 1353 \text{ W/m}^2 \text{ (in space)}$$

$$q''_{\text{sun, earth}} = 636 \text{ W/m}^2 \text{ (on the earth's surface)}$$

Note, the total arriving energy from sun to earth is $1.7 \times 10^{14} \text{ kW}$! Peak demand in the US is $1 \times 10^9 \text{ kW}$!

Most solar applications try to absorb some of the heat from the sun.

They use fancy surfaces called selective surfaces:



Because the absorption and emission spectra don't overlap, we can simplify our calculations:

$E_{\text{absorbed}} = q''_{\text{sun}} \cdot \alpha_1 \cdot A$	$\alpha_1 \equiv \text{absorptivity in the solar spectrum}$
$E_{\text{emitted}} = \epsilon_2 A \cdot \sigma T_{\text{surface}}^4$	$\epsilon_2 \equiv \text{emissivity in the IR spectrum}$